

ANA II - Woche 1

Def: (KURVEN): $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$

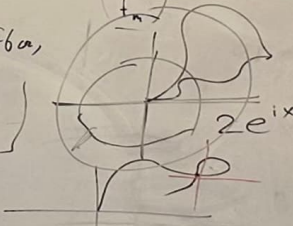
f_i $\begin{pmatrix} f_1 \\ f_2 \\ \vdots \\ f_n \end{pmatrix} \rightarrow f: \mathbb{R} \rightarrow \mathbb{R}$

$\gamma: I \rightarrow \mathbb{R}^n, I \subseteq \mathbb{R}, t \mapsto \gamma(t)$. Stetig diffbar,

$\dot{\gamma}(t) = \frac{d}{dt} \gamma(t)$
 Vektor $I := [t_0, t_1]$

$S(t) := \int_{t_0}^t \|\dot{\gamma}(t)\|_2 dt$

• geschlossen $\gamma(t_0) = \gamma(t_1)$
 • regulär $\dot{\gamma}(t) \neq 0 \forall t \in I$
 $t(\tau) \Rightarrow \gamma(t(\tau))$



1. $F = m \dot{r}$

2. Fkt. $\int f(z) dz$

$f \in C^1 \Rightarrow f' \in C^1$

$f \in C^1 \Rightarrow f \in C^\infty$

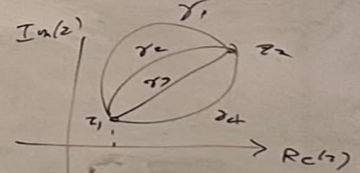
3. S.R.T. Weltlinie

Eigenzeit τ parametrisiert.

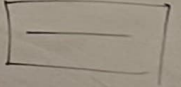
4. A.R.T.

$\begin{pmatrix} ct \\ \vec{r} \end{pmatrix}$

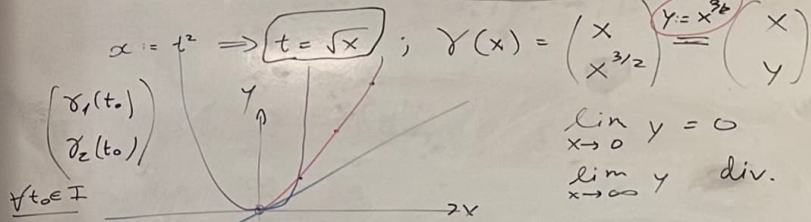
$f: U \rightarrow \mathbb{C}_1$
 $u \in \mathbb{C}_1$



$\mathbb{C}_1 \cong \mathbb{R}^2$
 CR.D.



P1.1: $\gamma: \mathbb{R}^+ \rightarrow \mathbb{R}^2, t \mapsto \gamma(t) := \begin{pmatrix} t^2 \\ t^3 \end{pmatrix}$



$\dot{\gamma}(t) = \begin{pmatrix} 2t \\ 3t^2 \end{pmatrix} \Rightarrow \|\dot{\gamma}(t)\| = \sqrt{(2t)^2 + (3t^2)^2} = t \sqrt{4 + 9t^2}$

$S(t) = \int_0^t \|\dot{\gamma}(t')\| dt' = \int_0^t t' \sqrt{4 + 9t'^2} dt' = \frac{1}{18} \int_0^t \sqrt{4 + 9t'^2} d(4 + 9t'^2)$

$= \left[\frac{1}{18 \cdot \frac{3}{2}} (4 + 9t'^2)^{3/2} \right]_0^t = \left[\frac{1}{27} \left((4 + 9t^2)^{3/2} - 8 \right) \right]$

$y = x^{3/2}, x > 0$

$\tilde{t}(s) = \sqrt{\frac{(27s + 8)^{2/3} - 4}{9}}$

$\gamma(\tilde{t}(s)) = \gamma(s) = \begin{pmatrix} \frac{(27s + 8)^{2/3} - 4}{9} \\ \left(\frac{(27s + 8)^{2/3} - 4}{9} \right)^{3/2} \end{pmatrix}$

$\rightarrow$ Monotonie der Integrand
 $E(m) = \lim_{\varepsilon \rightarrow 1^-} \int_0^\varepsilon \frac{\sqrt{1 - mt^2}}{\sqrt{1 - t^2}} dt$

P1.2: $E(m) := \int_0^1 \frac{\sqrt{1 - mt^2}}{\sqrt{1 - t^2}} dt = \lim_{\substack{\varepsilon \rightarrow 1^- \\ \varepsilon > 10}} \int_0^\varepsilon \frac{\sqrt{1 - mt^2}}{\sqrt{1 - t^2}} dt$
 $m \in [0, 1]$
 $t \in [0, 1]$

Spezialfälle: $m = 1: E(1) = \lim_{\varepsilon \rightarrow 1^-} \int_0^\varepsilon dt = 1$

$m = 0 \Rightarrow E(0) = \lim_{\varepsilon \rightarrow 1^-} \int_0^\varepsilon \frac{1}{\sqrt{1 - t^2}} dt = \lim_{\varepsilon \rightarrow 1^-} (\arcsin(\varepsilon) - \arcsin(0)) = \frac{\pi}{2} > 1$

$0 < \sqrt{1 - mt^2} \leq 1 \quad \forall m \in [0, 1] \Rightarrow 0 < \frac{\sqrt{1 - mt^2}}{\sqrt{1 - t^2}} \leq \frac{1}{\sqrt{1 - t^2}}$
 $\forall t \in (0, 1)$

P 1.2.
(b)

1. Parametrisierung

$$\gamma: [0, 2\pi] \rightarrow \mathbb{R}^2 \quad t \mapsto \gamma(t) = \begin{pmatrix} a \cos t \\ b \sin t \end{pmatrix}$$

$$s := L(\gamma) = \int_0^{2\pi} \|\dot{\gamma}(t)\| dt = \int_0^{2\pi} \sqrt{a^2 \sin^2 t + b^2 \cos^2 t} dt$$

$$= 4 \int_0^1 \sqrt{a^2 u^2 + b^2 (1-u^2)} \frac{du}{\sqrt{1-u^2}} = 4b \int_0^1 \frac{\sqrt{1 - u^2 \left(\frac{b^2 - a^2}{b^2}\right)}}{\sqrt{1-u^2}} du$$

$$= 4b \cdot \text{Elliptic } E\left(\frac{b^2 - a^2}{b^2}\right)$$

$$\dot{\gamma}(t) = \begin{pmatrix} -a \sin t \\ b \cos t \end{pmatrix}$$

$$\|\cdot\| = \sqrt{a^2 \sin^2 t + b^2 \cos^2 t}$$

$$u = \sin t \rightarrow \cos t dt = du$$

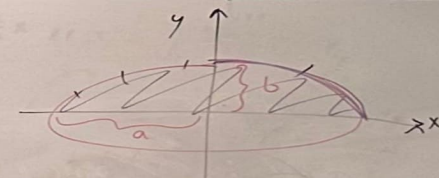
$$1 - u^2 = \cos^2 t \Rightarrow dt = \frac{du}{\sqrt{1-u^2}}$$

$$\begin{matrix} t=0 \rightarrow u=0 \\ t=\pi/2 \rightarrow u=1 \end{matrix}$$

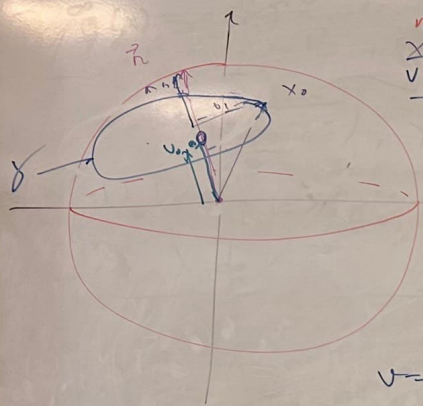
$$m = \frac{b^2 - a^2}{b^2}$$

$$r(t) =$$

$$\begin{pmatrix} r \sin \varphi \\ r \cos \varphi \end{pmatrix}$$



$$E(m) = \int_0^1 \frac{\sqrt{1-mt^2}}{\sqrt{1-t^2}} dt$$



$$\begin{array}{l} n \in S^1 \\ x_0 \in \mathbb{R}^3 \\ v \in \mathbb{R}^3 \end{array}$$

$$v = \langle n, x \rangle n$$

$$= x_0 - v$$

$$b_2 = n \times b_1 = n \times x_0$$

$$\Rightarrow \underline{b_1, b_2}$$

$$\gamma(t) = (x_0 \cdot n)n + \cos t (x_0 - (x_0 \cdot n)n) + \sin t (n \times x_0), t \in [0, 2\pi]$$

$$v = \langle n, x_0 \rangle n, b_1 = x_0 - v$$

P1.3

$$(a) \gamma: [0, 2\pi] \rightarrow \mathbb{R}^3,$$

$$t \mapsto \gamma(t) := c_1 b_1 + c_2 b_2 + c_3 \underbrace{S^2/2\text{-Sphäre}}_{\substack{b_1, b_2 \in \text{Ebene} \\ \text{Mittelpkt}}}$$

$$= r \cos t b_1 + r \sin t b_2 + v$$

$$\dot{\gamma}(t) = -r \sin t b_1 + r \cos t b_2$$

$$\ddot{\gamma}(t) = -r \cos t b_1 - r \sin t b_2$$

